



**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**  
**General Certificate of Education Ordinary Level**

**COMBINED SCIENCE**  
PAPER 1 Multiple Choice

**4003/1**  
1 hour

**SPECIMEN PAPER**

Additional materials:  
Multiple Choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)  
Calculator (Optional)

**INSTRUCTIONS TO CANDIDATES**

Answer **all** questions.

For each question, there are **four** possible answers, **A, B, C** and **D**. Choose the correct answer.

Record your choice in **soft pencil** on the separate answer sheet provided.

**Read very carefully the instructions on the answer sheet.**

**INFORMATION FOR CANDIDATES**

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

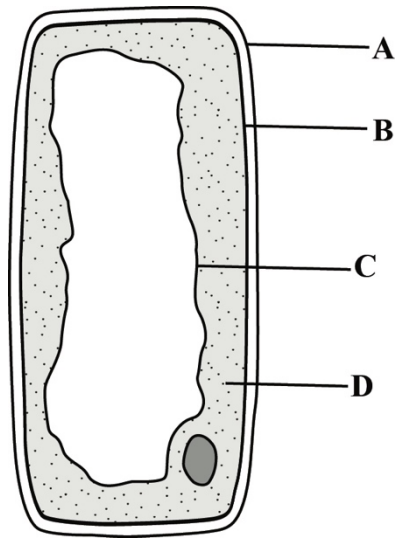
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**This specimen paper consists of 15 printed pages.**

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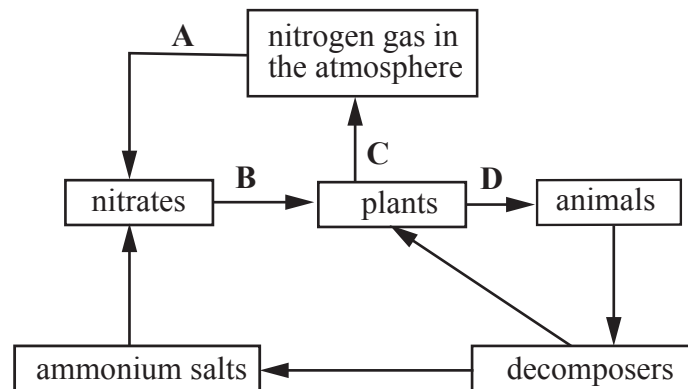
- 1 The diagram shows a plant cell.

Which part **A**, **B**, **C** or **D**, controls the movement of substances in and out of the cell?



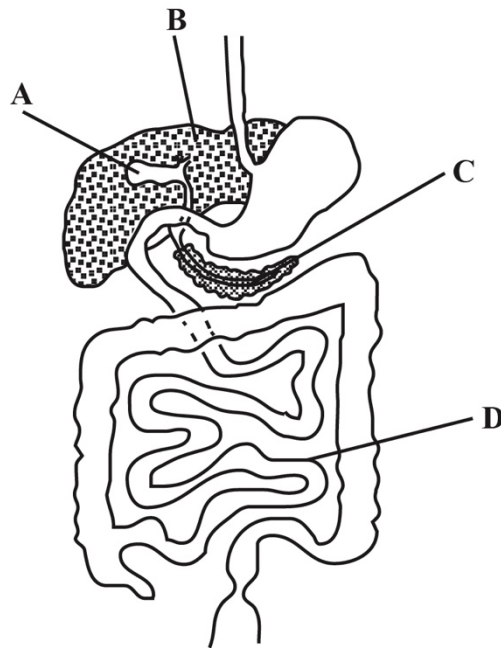
- 2 The diagram shows part of the nitrogen cycle.

Which process, **A**, **B**, **C** or **D**, results in increased soil fertility for sustainable food production?



- 3 The diagram shows part of the human digestive system.

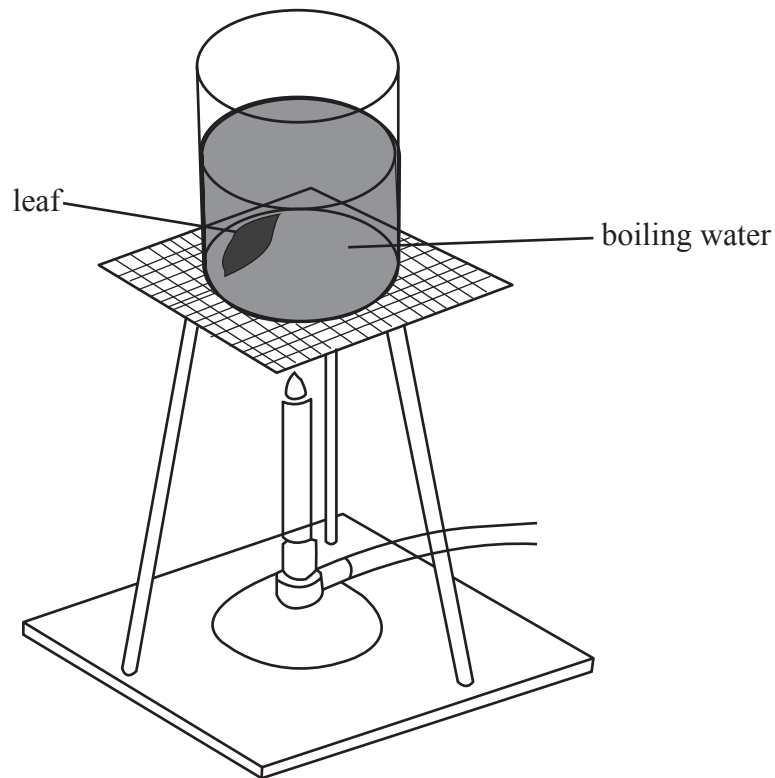
Which part, **A**, **B**, **C** or **D**, stores bile?



- 4 Which teeth are used for gripping and tearing food?

- A**     molars
- B**     incisors
- C**     canines
- D**     premolars

- 5 The diagram shows a stage in testing for starch in a leaf.



The stage is carried out to

- A soften the leaf.
  - B remove chlorophyll.
  - C show the presence of starch.
  - D kill the leaf cells.
- 6 Exhaled air has
- A more oxygen than inhaled air.
  - B less nitrogen than inhaled air.
  - C less water vapour than inhaled air.
  - D more carbon dioxide than inhaled air.
- 7 The equation for anaerobic respiration in muscle cells is



The product R represents

- A water.
- B energy.
- C alcohol.
- D carbon dioxide.

8 Which one is a function of the white blood cells?

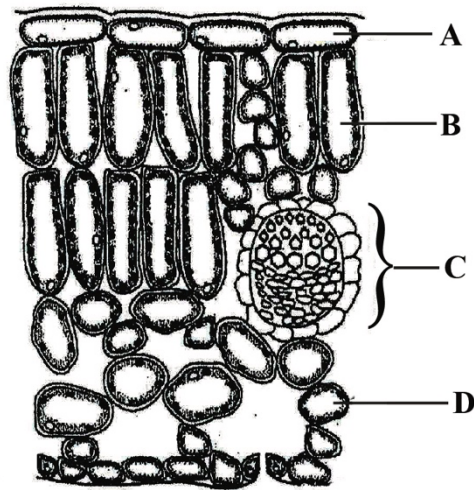
- A defence
- B transport
- C homeostasis
- D blood clotting

9 Transpiration is the movement of

- A water from the soil into the root.
- B water vapour from a leaf to the atmosphere.
- C carbon dioxide from the leaf to the atmosphere.
- D dissolved mineral ions from the soil into the root.

10 The diagram shows the cross section of a leaf.

Which part, A, B, C or D, is the vascular tissue?



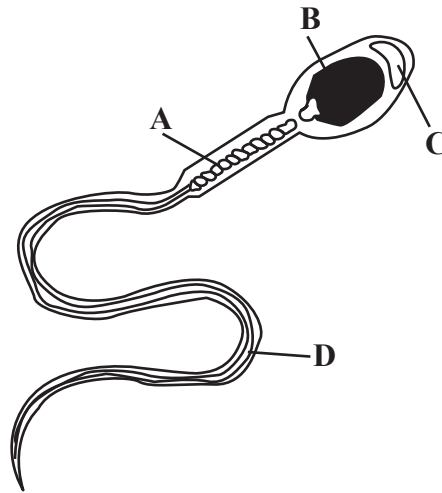
11 A farmer planted 50 bean seeds on a piece of land. 37 of the seeds germinated.

What is the percentage germination?

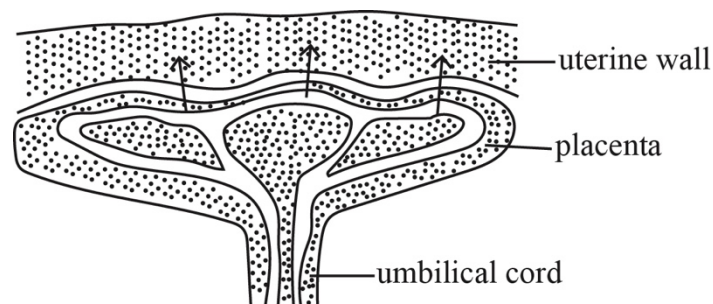
- A 13.0%
- B 26%
- C 74.0%
- D 87.0%

- 12 The diagram shows the male sex cell.

Which part contains enzymes that dissolve the cell membrane of an ovum?



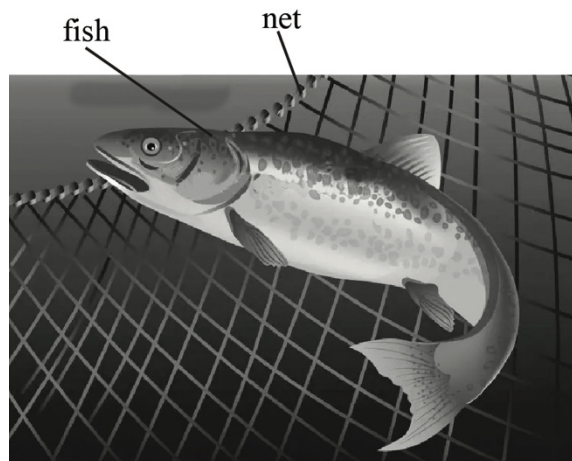
- 13 The diagram shows the movement of substances between the placenta and uterine wall.



The substance which moves in the direction of the arrows is

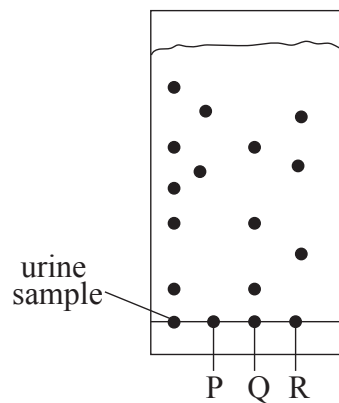
- A urea.
  - B oxygen.
  - C nutrients.
  - D antibodies.
- 14 Which one is a natural contraceptive method?
- A rhythm
  - B tubal ligation
  - C use of condoms
  - D oral contraceptive pill

- 15 Nets can be used for fishing by licensed fishermen.



This method is an application of

- A decanting.
  - B filtration.
  - C winnowing.
  - D evaporation.
- 16 The urine chromatogram of an athlete suspected to be taking performance enhancing drugs P, Q and R is shown. Samples of P, Q and R were compared with the urine sample.

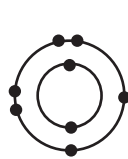


How many suspected drugs were in the urine sample?

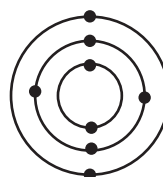
- A 0
- B 1
- C 2
- D 3

- 17 The nuclide notation for an oxygen atom is  $^{16}_8\text{O}$ .

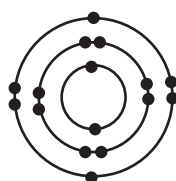
Which diagram shows the correct arrangement of electrons in the oxygen atom?



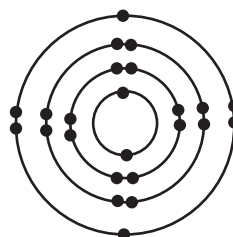
A



B



C



D

- 18 A  $0.025 \text{ dm}^3$  solution of sodium hydroxide ( $\text{NaOH}$ ) contains 0.025 moles.

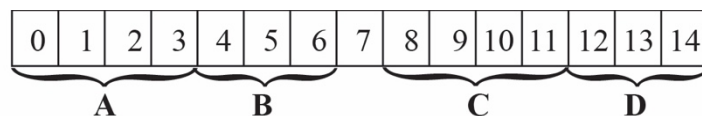
What is the concentration of the solution?

- A  $1.00 \text{ mol/dm}^3$ .  
 B  $0.075 \text{ mol/dm}^3$ .  
 C  $0.025 \text{ mol/dm}^3$ .  
 D  $0.00125 \text{ mol/dm}^3$ .
- 19 A hydrocarbon contains 6.25 moles of carbon (C) and 25.0 moles of hydrogen (H).  
 [Ar of C is 12; Ar of H is 1]

The empirical formula of the hydrocarbon is

- A  $\text{CH}_2$ .  
 B  $\text{CH}_4$ .  
 C  $\text{C}_2\text{H}_4$ .  
 D  $\text{C}_2\text{H}_6$ .
- 20 The diagram shows a pH scale.

Which range, A, B, C or D, shows the pH of lemon juice which is a weak acid?



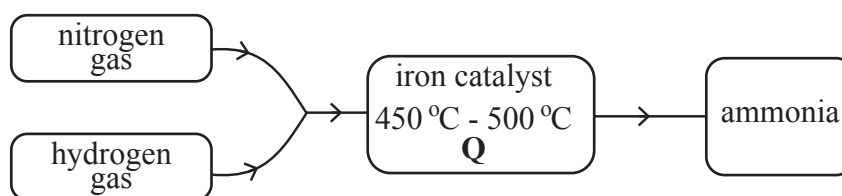


- 21 The table shows the observations made when metals **X**, **Y** and **Z** reacted with dilute hydrochloric acid.

| Metal    | Description of reaction |
|----------|-------------------------|
| <b>X</b> | reacts mildly           |
| <b>Y</b> | does not react          |
| <b>Z</b> | reacts violently        |

Which row correctly gives the metals in order of decreasing reactivity?

- A**    **X**     $\rightarrow$     **Y**     $\rightarrow$     **Z**  
**B**    **Y**     $\rightarrow$     **X**     $\rightarrow$     **Z**  
**C**    **Z**     $\rightarrow$     **X**     $\rightarrow$     **Y**  
**D**    **X**     $\rightarrow$     **Z**     $\rightarrow$     **Y**
- 22 The flow chart summarises the Haber process for the production of ammonia..



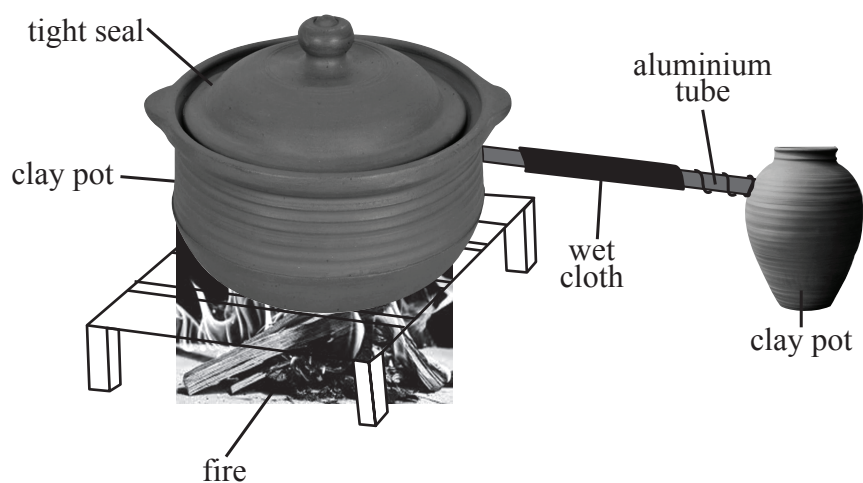
Condition **Q** is

- A**    200 atm.  
**B**    400 atm.  
**C**    500 atm.  
**D**    800 atm.
- 23 The function of limestone in the blast furnace is to
- A**    remove iron.  
**B**    reduce iron oxide.  
**C**    remove impurities.  
**D**    reduce calcium oxide.
- 24 Slag and iron, which are both liquids, do not mix.

This is because slag and iron have

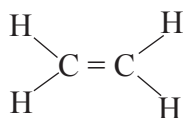
- A**    different densities.  
**B**    different melting points.  
**C**    different boiling points.  
**D**    different solubilities.

- 25** A mixture of water, brown sugar and yeast was placed in a warm place for four days. The mixture was then further processed as shown in the diagram.

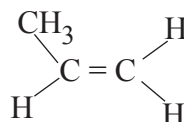


What is the purpose of the wet cloth?

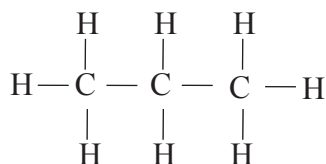
- A** to deliver alcohol vapour  
**B** to cool alcohol vapour  
**C** to maintain a constant temperature  
**D** to decorate the aluminium tube
- 26** The diagram shows four organic compounds.
- Which compound, **A**, **B**, **C** or **D**, is propene?



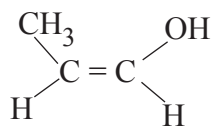
**A**



**B**



**C**



**D**

27 The burning of firewood is bad for the ecosystem because it leads to

- A a cool ecosystem.
- B the greenhouse effect.
- C more oxygen in the atmosphere.
- D less carbon dioxide in the atmosphere.

28 A 3.0 kg box measures 1 m by 2 m by 3 m.

What is the density of the box?

- A  $18.0 \text{ kg/m}^3$
- B  $9.0 \text{ kg/m}^3$
- C  $2.0 \text{ kg/m}^3$
- D  $0.5 \text{ kg/m}^3$

29 The newton is expressed in base units as

- A  $\text{kg/m/s}^2$ .
- B  $\text{kg m/s}^2$ .
- C  $\text{kg m}^2/\text{s}^2$ .
- D  $\text{kg m/s}$ .

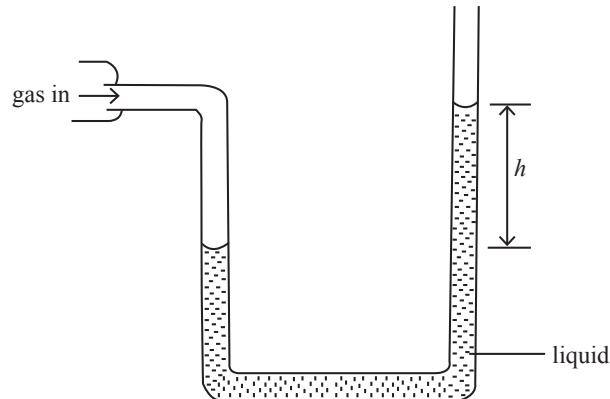
30 The diagram shows a simple machine.



The Mechanical Advantage (MA) of the machine is

- A 0.4.
- B 2.5.
- C 3.0.
- D 10.0.

- 31 The diagram shows an instrument used to measure gas pressure.



The instrument is a

- A barometer.
  - B manometer.
  - C forcemeter.
  - D micrometer.
- 32 A metal tin containing hot water and placed in a room cools faster if its outer wall is painted black rather than white.

This shows that black surfaces are better than white surfaces in

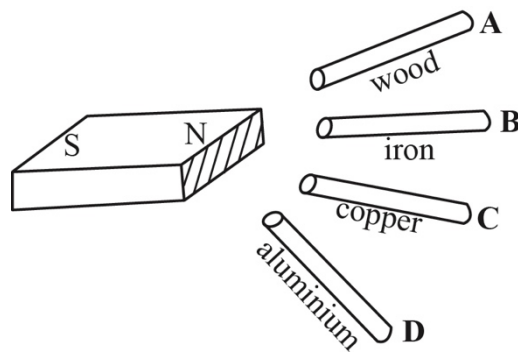
- A converting heat.
  - B conducting heat.
  - C reflecting heat.
  - D emitting heat.
- 33 Solar cells are used to supply electricity in homes.

Solar cells convert

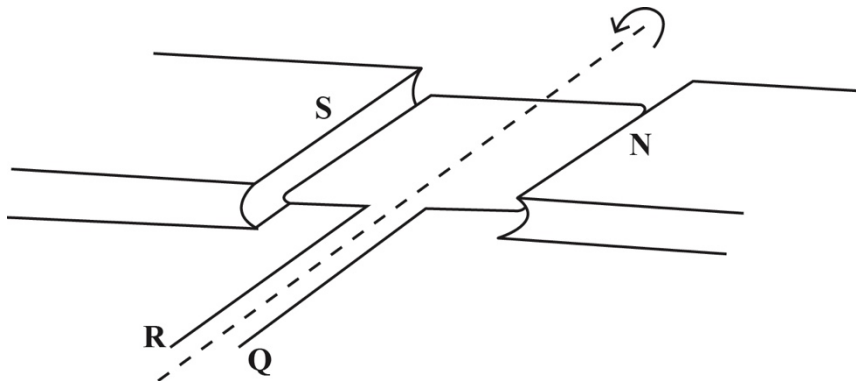
- A sunlight energy  $\rightarrow$  electrical energy.
- B chemical energy  $\rightarrow$  sunlight energy.
- C sunlight energy  $\rightarrow$  mechanical energy.
- D chemical energy  $\rightarrow$  mechanical energy.

- 34** The diagram shows four rods placed close to a magnet.

Which rod, **A**, **B**, **C** or **D**, is attracted to the magnet?



- 35** The diagram shows a coil in a magnetic field.



When an alternating current is generated, the coil is directly connected to **R** and **Q**.

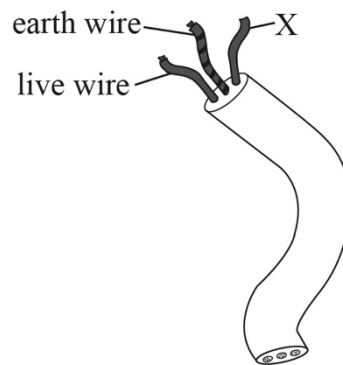
Parts **R** and **Q** are directly connected to the

- A** soft iron cores.
  - B** carbon brushes.
  - C** a.c. supplies.
  - D** slip rings.
- 36** A bulb which is connected to a 240.0 V supply draws a current of 0.3 A.

The resistance of the bulb is

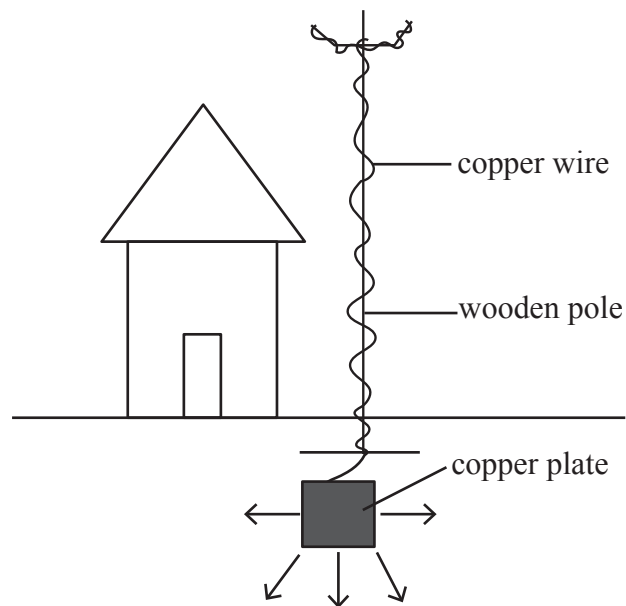
- A** 72.0  $\Omega$ .
- B** 239.7  $\Omega$ .
- C** 240.3  $\Omega$ .
- D** 800.0  $\Omega$ .

- 37 The diagram shows a cable used in a three pin-plug.



The colour code for wire X is

- A blue.
  - B brown.
  - C green.
  - D yellow.
- 38 A lightning conductor is used to protect a building as shown.



The purpose of the copper wire is to conduct

- A protons to the ground.
- B neutrons to the ground.
- C nucleons to the ground.
- D electrons to the ground.

**39** Which truth table is for an OR gate?

| input<br>X | input<br>Y | output<br>Z |
|------------|------------|-------------|
| 1          | 1          | 0           |
| 1          | 0          | 0           |
| 0          | 1          | 0           |
| 0          | 0          | 1           |

**A**

| input<br>X | input<br>Y | output<br>Z |
|------------|------------|-------------|
| 1          | 1          | 1           |
| 1          | 0          | 1           |
| 0          | 1          | 1           |
| 0          | 0          | 0           |

**B**

| input<br>X | input<br>Y | output<br>Z |
|------------|------------|-------------|
| 1          | 1          | 0           |
| 1          | 0          | 1           |
| 0          | 1          | 1           |
| 0          | 0          | 1           |

**C**

| input<br>X | input<br>Y | output<br>Z |
|------------|------------|-------------|
| 1          | 1          | 1           |
| 1          | 0          | 0           |
| 0          | 1          | 0           |
| 0          | 0          | 0           |

**D**

**40** Which one is an example of a sensor?

- A** pump
- B** valve
- C** motor
- D** thermistor